

1.

List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Three Real-Life Uses of a Multimeter

1. **Computer not powering on:** Check the **power supply voltage** to see whether the PSU is providing the correct voltage.
2. **Laptop charging problem:** Test the **charger/adaptor output voltage** to determine if the charger is working properly.
3. **Smartphone not charging:** Check for **voltage and continuity** in the charging port or cable to identify a faulty connection.

2.

Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

The POST card displayed **two-digit hexadecimal codes** while the motherboard performed its Power-On Self-Test (POST).

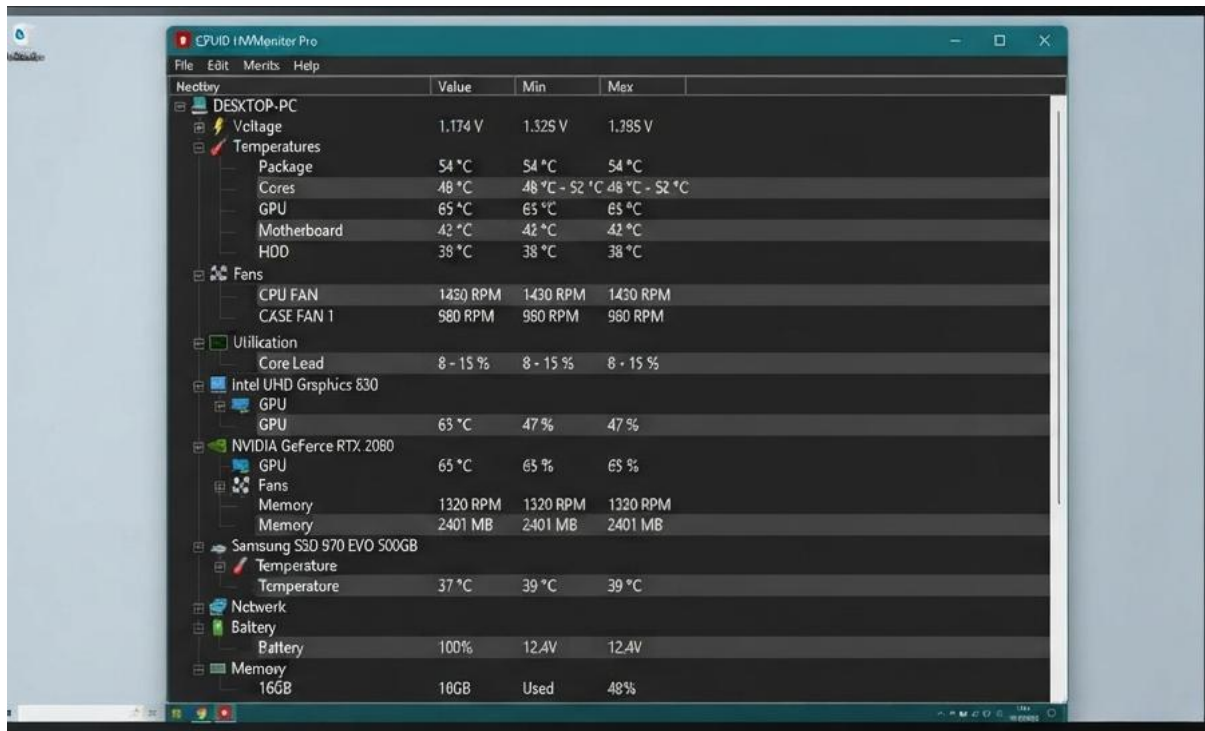
The codes changed as the BIOS tested different hardware components such as the CPU, RAM and other devices.

When the POST process stopped, the last displayed code helped indicate where the hardware problem was located.

This made it easier to troubleshoot a **computer that would not boot or showed a blank screen.**

3.

Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.



4.

Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem Detected
Multimeter	Checks electrical voltage, resistance, and continuity.	Faulty power supply/charger or broken cable connection.
POST Card	Checks motherboard Power-On Self-Test (POST) codes during startup.	RAM or CPU problem preventing the computer from booting.
Software Diagnostics	Monitors system health such as temperature, storage health, and hardware status.	Overheating CPU or a failing hard drive/SSD.